Regd. No. | | | | | | | | | | |

VIGNAN'S
FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH
(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

Regulation: R22C24 **Course Code: 24PY102-5**

I B.Tech I Semester Regular Examinations – January, 2025

ENGINEERING PHYSICS

(CHEM/CIVIL/CS&BS/CSE(CS)/CSE(DS)/CSE(AI&ML)/CSE(IOT)/
ECE/ECE(VLSI)/EEE/IT/MECH/TT)

Time: 150 Min **Max. Marks:60M**

SECTION – A

Answer all Four questions **4×8M=32M**

1. a) Outline the differences between Direct and Indirect bandgap semiconductors with neat band diagrams. **(6M)**
   b) Why do mobile phones and computers get heated after usage over some time?**(1M)**
   c) Why do you prefer direct bandgap semiconductors rather than indirect bandgap semiconductors in light-emitting devices such as LEDs and LASERs? **(1M)**

2. a) Explain how the stimulated emission helps to obtain the properties needed for LASER. **(2M)**
   b) What are the essential conditions to obtain a LASER? **(2M)**
   c) With a neat sketch elucidate the working of semiconductor LASER. **(4M)**

3. a) What is Heisenberg's uncertainty principle? Explain it with a suitable example. **(4M)**
   b) Is Heisenberg's uncertainty principle valid for macroscopic particles? **(1M)**
   c) The particle is confined in a potential well of length 1A°. The position and momentum of an electron with energy of 1keV are simultaneously measured. Calculate the percentage error in its momentum. **(3M)**

4. a) What is the Hall Effect? **(1M)**
   b) Explain how the Hall Effect is useful to distinguish n-type and p-type semiconductors with necessary mathematical expressions. Write the mathematical expressions to extract the parameters such as mobility and conductivity using Hall Effect. **(5M)**
   c) A semiconductor has the dimensions of 5 mm length, 5 mm wide and a thickness of 1mm and is placed in a transverse magnetic field of 0.5T when a current of 20 A flows lengthwise through the specimen. Calculate the carrier concentration, if the Hall voltage measured across the sample is 20μA. **(2M)**

SECTION - B

Answer all Two questions **2×14M=28M**

5. a) Why doping is necessary in semiconductors? How electron and hole conductivity can be controlled with doping? **(7M)**
   b) Calculate the resistance of a pure silicon crystal of length 1cm and $4cm^2$ area of cross section and having an intrinsic carrier concentration of $2.5\times10^{16}/m^3$. The electron and hole mobilities are $1400cm^2/Vs$ and $500cm^2/Vs$ respectively. Also, calculate the resistance and the majority and minority carrier concentration when the material is doped with 10^{21} Phosphorus atoms. **(3M)**
   c) What is the principle of LED? Elucidate the working of LED with the help of a band diagram. **(4M)**

6. a) What are the assumptions of the classical free electron theory? **(3M)**
   b) Apply the Newton's laws to obtain an expression for the electrical conductivity and current density of a metal. **(6M)**
   c) What is the Fermi-Dirac distribution function? Explain its variation with temperature and its significance at T=0K and T>0K. **(4M)**
   d) Plot the graph explaining the variation of the Fermi-Dirac distribution function at different temperatures. **(1M)**